

Tewodros Kederalah Idris

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PROFILE

Master’s graduate in Engineering Artificial Intelligence with hands-on experience designing, training, and deploying machine learning systems using Python, PyTorch, and Hugging Face Transformers. Strong background in NLP, retrieval-augmented generation (RAG), and multimodal reasoning. Skilled at translating advanced research into scalable, production-ready ML solutions through data-driven experimentation and automation.

EDUCATION

Carnegie Mellon University Africa <i>Master of Science, Engineering Artificial Intelligence; GPA: 3.88</i> Coursework: Generative AI, Introduction to Deep Learning, Question Answering with LLMs	Kigali, Rwanda Aug 2023 – May 2025
10 Academy <i>Machine Learning & Data Engineering Bootcamp</i> Coursework: MLOps, data visualization, machine learning, data engineering	Remote May 2022 – Jul 2023
Addis Ababa University <i>Bachelor of Science, Computer Engineering; GPA: 3.69</i> Graduated with great distinction	Addis Ababa, Ethiopia Jan 2015 – Jan 2020

EMPLOYMENT HISTORY

Research Associate <i>Carnegie Mellon University Africa - Language and Spatial Technologies Lab</i> - Built scalable pipelines to measure mutual intelligibility across multilingual language models (AfriBERTa, LaBSE) using FLORES+ datasets covering 200+ languages. - Automated large-scale evaluation and visualization workflows with Python, PyTorch, and Hugging Face Transformers, streamlining experimentation and data analysis. - Developed a three-stage framework integrating embedding similarity analysis, monolingual controls, and URIEL+ typological validation to assess cross-lingual model behavior. - Introduced rank-based evaluation metrics (AUC) to improve robustness in multilingual representation analysis. - Analyzed tokenization patterns and embedding geometries to uncover linguistic factors driving model generalization in low-resource African languages.	Aug 2025 – Present Kigali, Rwanda
Machine Learning Research Assistant <i>Carnegie Mellon University - Language Technologies Institute</i> - Developed and fine-tuned Vision-Language Models (VLMs) for generating synthetic visual data, enhancing reasoning performance through visual context integration. - Extended existing research pipelines with state-of-the-art models using Hugging Face Transformers and PyTorch for efficient training and evaluation. - Conducted large-scale experiments on benchmarks (GSM8K, ASDiv, SVAMP) to assess reasoning consistency and model robustness. - Optimized model architectures by evaluating visual feedback mechanisms to improve multi-step reasoning capabilities. - Collaborated with faculty and researchers to analyze outcomes, identify failure cases, and contribute to publication-ready results.	Feb 2025 – Present Pittsburgh, US
Graduate Teaching Assistant <i>Carnegie Mellon University</i> - Led NLP coursework focused on Transformer-based architectures and practical applications in modern language modeling. - Created lab materials, guided student projects, and held weekly office hours to strengthen understanding of deep learning concepts.	Sep 2024 – Dec 2024 Pittsburgh, US

Research Assistant

May 2024 – Aug 2024

Carnegie Mellon University - Language Technologies Institute (WAVLab)

Pittsburgh, US

- Extended the speech processing toolkit to support low-resource languages (Amharic, Swahili), contributing to scalable ASR and retrieval pipelines.

Protocol Engineer

Mar 2021 – Aug 2023

NuNet

Remote

- Co-led the development of a decentralized peer-to-peer compute platform, improving scalability and reliability across distributed nodes.
- Built a device onboarding mechanism to enable cross-platform compatibility and seamless cluster integration.
- Implemented microservices-based architecture using Docker and Linux for efficient orchestration of compute tasks.
- Developed lightweight micro virtual machines to enable remote access for distributed GPU job deployment.

Machine Learning Intern

Mar 2019 – Feb 2021

iCog-Labs

Addis Ababa, Ethiopia

- Developed an Amharic language model using RNNs and LSTMs for a keyboard application, improving language coverage for local users.
- Integrated Amharic capabilities into Sophia the robot, expanding multilingual interaction and speech capabilities.
- Generated annotated image datasets with GANs to improve facial recognition models and training efficiency.

SKILLS & TOOLS

Programming: Python, Go, Bash, C++

Frameworks: PyTorch, Hugging Face Transformers, ESPNet, Scikit-learn

MLOps & Infrastructure: Docker, Git, Linux, Apache Kafka, SQL

Domains: NLP, Speech Recognition, RAG, Vision-Language Models, Multimodal Learning, Deep Learning, LLMs, Information Retrieval, Prompt Engineering

SELECTED PROJECTS & PUBLICATIONS

ColBERT Retrieval and Ensemble Response Scoring for Language Model Question Answering

2024

- Developed a retrieval-augmented QA pipeline that improved small language model accuracy through ColBERT-based retrieval and ensemble response scoring.
- Achieved accuracies of 81.9% (Phi-2) and 57.3% (Falcon-7B) on a domain-specific telecommunications QA benchmark.
- Published findings on enhancing domain-specialized reasoning through retrieval and response optimization (arxiv.org/abs/2408.10808).

AWARDS AND ACHIEVEMENTS

Specializing Large Language Models for Telecom Networks

ITU AI/ML in 5G Challenge

- Led model adaptation and retrieval optimization for a domain-specific QA system, contributing to a top-ranked global submission.
- Integrated response scoring and retrieval enhancements to improve inference quality in telecom knowledge systems.